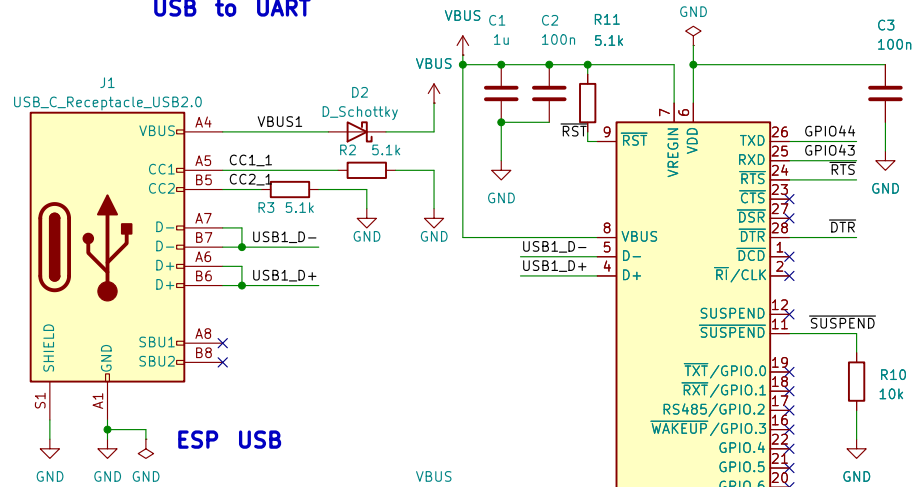


EasyDuino ESP32 S3

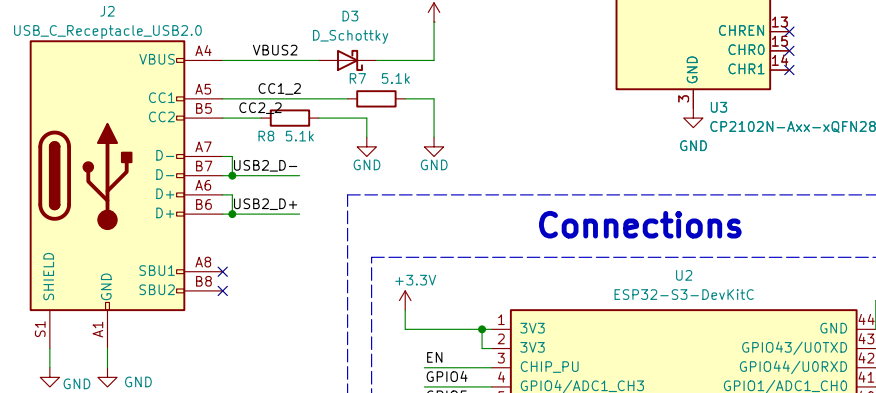
USB-C/Serial/Automatic Bootloader

ESP32 Module

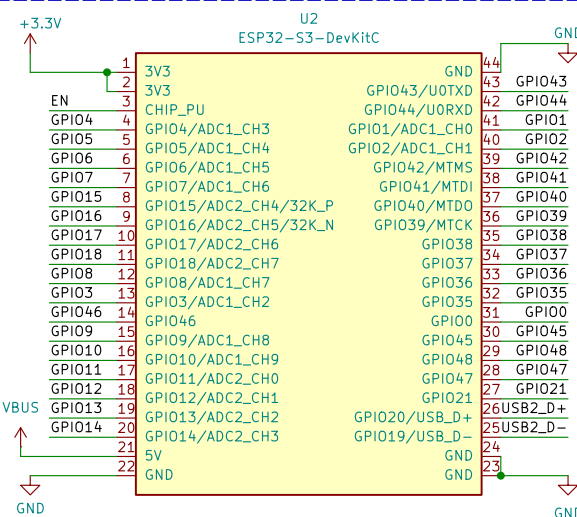
USB to UART



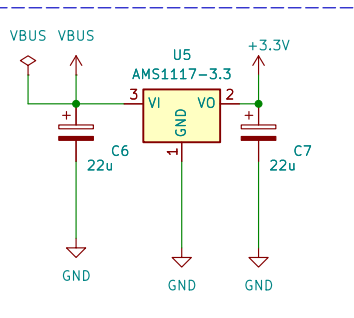
ESP USB



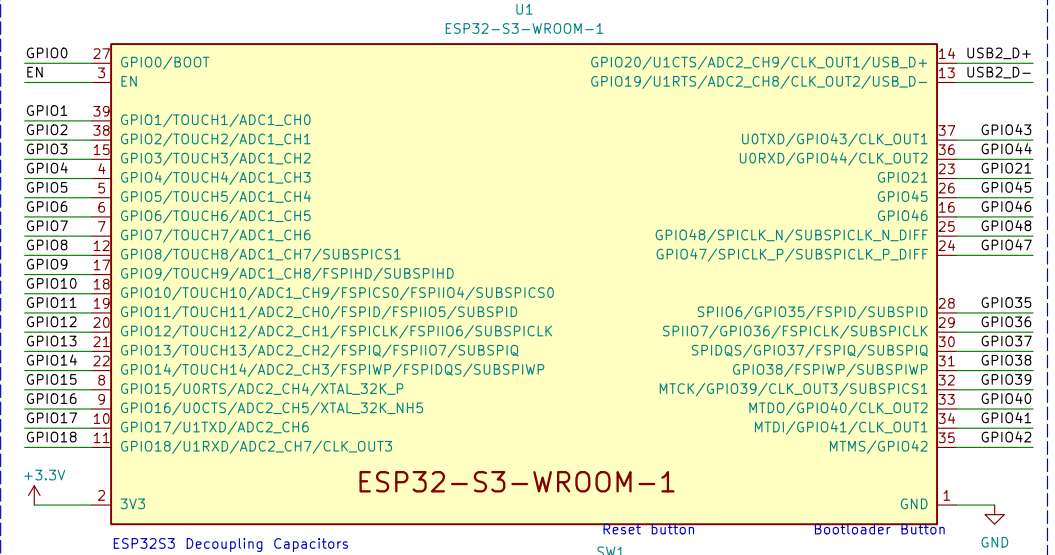
Connections



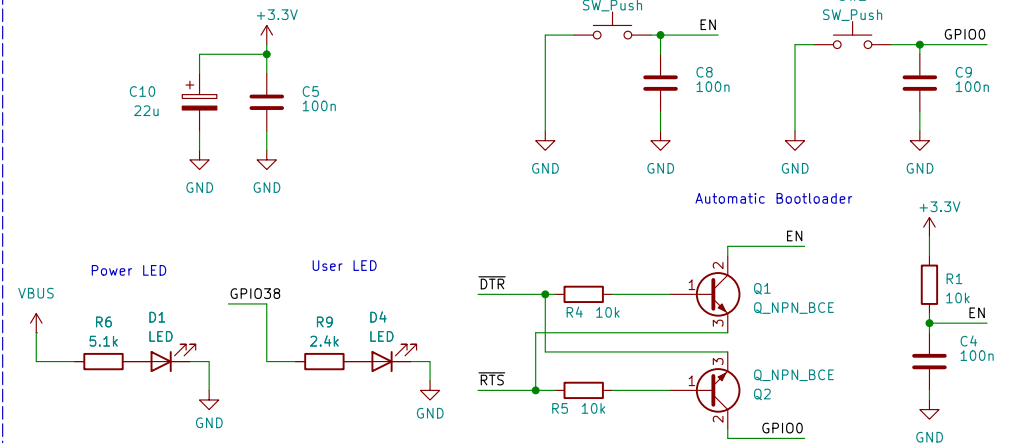
Voltage Regulator



ESP32-S3-WROOM-1



ESP32S3 Decoupling Capacitors



References

File: References.kicad_sch

Sheet: /
File: Easyduino_ESP32S3.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:
Id: 1/2

Specs:

Input Voltage: 5V–15V(External Source 5V) 5V (USB UART OR USB ESP32) or 3.3V. ONLY USE ONE VOLTAGE INPUT AT A TIME
Output Current at 3.3V : 1 A (External Source) 500mA (USB)
No Polarity Protection (External Source)
Overcurrent Protection: None

References:

ESP32 S3 WROOM 1

Decoupling Capacitors:

ESP32 S3 WROOM 1 datasheet p25 -> 3.3V pin needs a 22uF and a 100nF capacitors. The 22uF doesn't have to be a tantalum capacitor.
But I will use a tantalum to reuse components

Automatic Bootloader:

ESP32 Sch -> Automatic Bootloader.

ESP32 Sch / ESP32 S3 WROOM 1 datasheet p25 -> EN pin 10k resistor and 100nF capacitor

Thermal Vias:

ESP32 S3 WROOM 1 datasheet p27 -> GND Thermal needs vias between pads

USB C

Filtering:

USB Hardware Design Guidelines for FTDI ICs p8 -> 10nF Cap + Ferrite Bead + 100nF Cap + 4.7uF Cap
I disregarded such heavy filtering for the easier to implement 1uF and 100nF cap given in the CP2102 datasheet

Differential pair:

USB needs 90 Ohm differential traces.

Using JLCPCB's JLC04161H–7628 stackup.

Which with Signal Layer (Layer 1) and a GND layer (L2) below. And with 0.2mm tracks and separated 0.15mm gives a differential pair impedance of approximately 90 Ohm

For USB1 will use a very short trace for D+ D- (around 2mm)

For USB2 (tracks that go to the internal ESP32 S3 pins and the pinout) the traces are long ~60mm for D+ D-, and have vias

The Development board from ESP32S3 has no problem with these differential pairs with vias and is a 2 board layer

I also have to place vias because of the pinout of USB C and CP2102. Some other people have placed vias without problems for signal integrity

CP2102

CP2102 datasheet p15 -> VBUS will be tied to REGIN. 1uF and 100 nF will be added.

CP2102 datasheet p15 -> ESD Supression Diode I will be skipping it

CP2102 datasheet p15 and p19 -> 100nF for VDD output, 4.7 uF is optional for higher current circuits.

CP2102 datasheet p23 -> The VDD or 3.3 output can only provide 100mA. I will not use this as a voltage provider for the circuit. I will use the AMS1117–3.3

CP2102 datasheet p15 -> RST pulled up via 4.7k and SUSPEND~ pulled down via 10k.

AMS1117–3.3V

AMS1117 Datasheet p1 -> 3.3V/1A Output with 5–15V Input

AMS1117 Datasheet p4 -> 22uF tantalum capacitor on input and output.

Sheet: /References/
File: References.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 2/2